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Display device

Abstract

A display device including a display module and a support structure is provided. The support structure is connected to the display module and supports the display module. The support structure includes a support plate and a plastic frame. The support plate includes a first lateral surface and a second lateral surface. The plastic frame is protrusively disposed on the first lateral surface. The plastic frame is distributed at a periphery of the light transmissive region. By disposing the support structure and connecting the support structure to the display module, a structural strength of an entire display device is improved, which is a simple structure and is convenient for assembling.

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Background/Summary

RELATED APPLICATIONS

(1) This application is a National Phase of PCT Patent Application No. PCT/CN2021/094558 having International filing date of May 19, 2021, which claims the benefit of priority of Chinese Patent Application No. 202110470869.2 filed on Apr. 28, 2021. The contents of the above applications are all incorporated by reference as if fully set forth herein in their entirety.

FIELD AND BACKGROUND OF THE INVENTION

(2) The present application relates to the field of display technology and specifically relates to a display device.

(3) With development of liquid crystal display technology, applications of double-sided display devices have become more and more widespread. In double-sided display devices, two display panels are stacked and assembled and have advantages of ability to display images on two sides, which can be used in offices of communication industry, government services, financial industry, transportation industry, and other industries. When the double-sided display devices are used as televisions in daily life, they can be embedded in door frames or partition walls of adjacent rooms, so that people in different rooms can obtain information at a same time.

(4) In order to reduce a thickness of the double-sided display devices, one light guide plate with a light source and a transflective film acting as a backlight assembly is adopted in current double-sided display devices to provide backlight for two display panels. Each film layer in the display panel is usually selected as a relatively thin type, resulting in poor rigidity of the double-sided display devices and easy deformation, thereby causing damage to the double-sided display devices.

SUMMARY OF THE INVENTION

(5) The present application provided a display device, which aims to solve the problem of poor rigidity and easy deformation of the current double-sided display devices.

(6) On a first aspect, the present application provides a display device, the display device includes: a display module including a first display panel and a second display panel disposed oppositely and a backlight component disposed between the first display panel and the second display panel, wherein, the backlight component includes a light guide plate and a light source disposed on a lateral side of the light guide plate, the first display panel and the second display panel have light incident surfaces, and the light incident surfaces of the first display panel and the second display panel are respectively opposite to the backlight component; a support structure, wherein the support structure is connected to the display module and supports the display module, the support structure includes a support plate and a plastic frame, the support plate includes a first lateral surface and a second lateral surface which are opposite to each other, the support plate has a light transmissive region, the plastic frame is protrusively disposed on the first lateral surface, the plastic frame is distributed at a periphery of the light transmissive region, and a surface of the plastic frame away from the support plate is connected to the first display panel.

(7) Optionally, the support structure further includes a lateral wall protrusively disposed on a lateral edge of the support plate, the support plate and the lateral wall enclose an accommodating space, the backlight component is disposed in the accommodating space; a support part extends from a side of the lateral wall away from the support plate toward the accommodating space, and a surface of the support part away from the support plate is connected to the second display panel.

(8) Optionally, a via hole is defined on the support plate to form the light transmissive region, and the plastic frame is disposed on a lateral edge of the via hole.

- (9) Optionally, a boss is disposed on the first lateral surface close to the lateral edge of the via hole, a groove corresponding to the boss is defined on the plastic frame, and the boss is inserted in the groove.
- (10) Optionally, the backlight component includes a light guide plate and a light source disposed on a side of the light guide plate, the lateral wall comprises a first lateral wall and a second lateral wall sequentially distributed from the light source to the light guide plate, the support part extends from a side of the first lateral wall away from the support plate toward the accommodating space, and the surface of the support part away from the support plate is connected to the second display panel.
- (11) Optionally, a surface of the support part close to the support plate is connected to the light guide plate.
- (12) On a second aspect, the present application provides a display device, the display device includes: a display module including a first display panel and a second display panel disposed oppositely and a backlight component disposed between the first display panel and the second display panel, wherein the first display panel and the second display panel have light incident surfaces, and the light incident surfaces of the first display panel and the second display panel are respectively opposite to the backlight component; a support structure, wherein the support structure is connected to the display module and supports the display module, the support structure includes a support plate and a plastic frame, the support plate includes a first lateral surface and a second lateral surface which are opposite to each other, the support plate has a light transmissive region, the plastic frame is protrusively disposed on the first lateral surface, the plastic frame is distributed on a periphery of the light transmissive region, and a surface of the plastic frame away from the support plate is connected to the first display panel.
- (13) Optionally, the support structure further includes a lateral wall protrusively disposed on a lateral edge of the support plate, the support plate and the lateral wall enclose an accommodating space, the backlight component is disposed in the accommodating space; a support part extends from a side of the lateral wall away from the support plate toward the accommodating space, and a surface of the support part away from the support plate is connected to the second display panel.
- (14) Optionally, a via hole is defined on the support plate to form the light transmissive region, and the plastic frame is disposed on a lateral edge of the via hole.
- (15) Optionally, a boss is disposed on the first lateral surface close to the lateral edge of the via hole, a groove corresponding to the boss is defined on the plastic frame, and the boss is inserted in the groove.
- (16) Optionally, the backlight component includes a light guide plate and a light source disposed on a side of the light guide plate, the lateral wall includes a first lateral wall and a second lateral wall sequentially distributed from the light source to the light guide plate, the support part extends from a side of the first lateral away from the support plate toward the accommodating space, and the surface of the support part away from the support plate is connected to the second display panel.
- (17) Optionally, a surface of the support part close to the support plate is connected to the light guide plate.
- (18) Optionally, a connection element is disposed on the support plate close to the second lateral wall, and an end of the connection element is connected to the second display panel, and another end of the connection element is connected to the support plate.
- (19) Optionally, an area of the first display panel is smaller than an area of the second display panel.
- (20) Optionally, a material of the support plate is stainless steel or aluminum.
- (21) Optionally, the support plate and the plastic frame are integrally formed.
- (22) Optionally, the plastic frame and the support plate enclose to form a mounting cavity, the second display panel is disposed in the mounting cavity, and the second display panel is located between the first display panel and the first lateral surface and is connected to the first lateral surface.

- (23) Optionally, a material of the support plate is resin or glass.
- (24) Optionally, a lateral side of the plastic frame is connected to a lateral side of the backlight component.
- (25) Optionally, an area of the second display panel is less than or equal to an area of the first display panel.
- (26) The present application provides a display device, including a display component and a support structure. The display module includes the first display panel and the second display panel disposed oppositely and the backlight component disposed between the first display panel and the second display panel. The first display panel and the second display panel have light incident surfaces. The light incident surfaces of the first display panel and the second display panel are respectively opposite to the backlight component. The support structure is connected to the display module and supports the display module. The support structure includes the support plate and the plastic frame. The support plate includes the first lateral surface and a second lateral surface which are opposite to each other. The support plate has a light transmissive region. The plastic frame is protrusively disposed on the first lateral surface. The plastic frame is distributed on a periphery of the light transmissive region. The surface of the plastic frame away from the support plate is connected to the first display panel. By disposing the support structure and connecting the support structure to the first display panel and the second display panel, the structural strength of the entire display device is improved, preventing deformation from occurring. Furthermore, the structure is relatively simple and is conducive to being assembled.
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Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

- (1) The technical solutions and other advantageous effects of the present invention will be apparent with reference to the following accompanying drawings and detailed description of embodiments of the present disclosure.
- (2) FIG. 1 is structural schematic diagram of a display device of a first embodiment of the present application.
- (3) FIG. 2 is a structural schematic diagram of the display device of a second embodiment of the present application.
- (4) FIG. 3 is a structural schematic diagram of the display device of a third embodiment of the present application.
- (5) FIG. 4 is a structural schematic diagram of the display device of a fourth embodiment of the present application
- (6) FIG. 5 is a top view of the display device in FIG. 4.
- (7) FIG. 6 is a structural schematic diagram of the display device of a fifth embodiment of the present application.
- (8) TABLE-US-00001 reference reference numbers in numbers in accompanying name of accompanying name of figures element figures element 1 display device 10 display module 11 first display 211 connection panel element 12 second display 13 backlight panel component 134 first optical 135 second optical film layer film layer 100 light incident 131 light surface source 132 light guide plate 14 support structure 133 transfective film d distance 141 support plate 142 plastic frame 1411 first lateral 1412 second lateral surface surface 140 transmissive region 1413 lateral wall 1413a first lateral 1413b second lateral wall wall 150 accommodating 1414 support space part 1400 via hole 1410 boss 1420 groove 200 mounting cavity
- ### DESCRIPTION OF SPECIFIC EMBODIMENTS OF THE INVENTION
- (9) The technical solutions in the embodiments of the present disclosure are clearly and completely described in the following with reference to the accompanying drawings in the embodiments of the

present disclosure. Obviously, the described embodiments are only part of the embodiments of the present disclosure, and are not all embodiments of the present disclosure. All other embodiments obtained by those skilled in the art based on the embodiments of the present disclosure without creative efforts are within the scope of the present disclosure. Besides, it should be understood that the specific embodiments described herein are merely for describing and explaining the present disclosure and are not intended to limit the present disclosure. In the present disclosure, unless opposite stated, the orientation words used such as “upper” and “lower” generally refer to the upper and lower directions of the device in actual using or working state, and specifically refer to the drawing directions in the drawings, and “inner” and “outer” refer to the outline of the device.

(10) One embodiment of the present application provides a display device. The details are described below respectively. It should be noted that a description order of the following embodiments is not intended to limit a preferred order of the embodiments.

(11) FIG. 1 is structural schematic diagram of a display device **1** of a first embodiment of the present application. As illustrated in the figure, the display device **1** includes a display module **10** and a support structure **14**. The display module **10** includes a first display panel **11** and a second display panel **12** disposed oppositely and a backlight component **13** disposed between the first display panel **11** and the second display panel **12**. The first display panel **11** and the second display panel **12** have light incident surfaces **100**. The light incident surfaces **100** of the first display panel **11** and the second display panel **12** are respectively opposite to the backlight component **13**. As such, the backlight component **13** provides a light source **131** to the first display panel **11** and the second display panel **12**.

(12) The support structure **14** includes a support plate **141** and a plastic frame **142**. The support plate **141** includes a first lateral surface **1411** and a second lateral surface **1412** which are opposite to each other. The support plate **141** has a light transmissive region **140**. The light transmissive region **140** corresponding to a first optical film layer **134** of the backlight component **13** allows light emitted from the backlight component **13** to pass through the support plate **141** and to irradiate on the first display panel **11**. The plastic frame **142** is protrusively disposed on the first lateral surface **1411** and is distributed on a periphery of the light transmissive region **140**. The surface of the plastic frame **142** away from the support plate **141** is connected to the first display panel **11** to support the first display panel **11**.

(13) In the first embodiment of the present application, the second display panel **12** is opposite to the second lateral surface **1412** and is connected to the support plate **141**.

(14) As illustrated in FIG. 1, the support structure **14** further includes a lateral wall **1413** protrusively disposed on a lateral edge of the support plate **141**; the support plate **141** and the lateral wall **1413** enclose an accommodating space **150**; the backlight component **13** is disposed in the accommodating space **150**; a support part **1414** extends from a side of the lateral wall **1413** away from the support plate **141** toward the accommodating space **150**, i.e., the support part **141** is formed from extension of the lateral wall **1413**; and a surface of the support part **1414** away from the support plate **141** is connected to the second display panel **12** to support the second display panel **12**.

(15) In order to reduce a thickness and a weight of the display module **10** to realize thinness, the first display panel **11** and the second display panel **12** are provided with the light source **131** by a layer of the backlight component, and a relatively thin film material is also used on the first display panel **11** and the second display panel **12**. As a result, the display module **10** composed of the first display panel **11**, the second display panel **12**, and the backlight component **13** has poor rigidity and is prone to be deformed, causing damage to the display module **10**.

(16) By disposing the support structure **14** including the support plate **141** and the plastic frame **142**, by supporting the first display panel **11** through the plastic frame **142**, and by supporting the second display panel **12** through the support plate **141**, connection between the support structure **14** and the display module **10** and support for the display module **10** from the support structure **14** are

realized, which improves rigidity of the display module **10** and prevents the display module **10** from being deformed. In addition, the support structure **14** is composed of the support plate **141** and the plastic frame **142** protrusively disposed on the support plate **141**, which is a simple structure convenient to fabricate, and is also conducive to assembling the display device **1**.

(17) In the present application, the backlight component **13** includes a light guide plate **132**, the light source **131** disposed on a side of the light guide plate **132**, a transfective film **133** disposed between the light guide plate **132** and the support plate **141**, and a first optical film layer **134** and a second optical film layer **135** respectively disposed on two sides of the light guide plate **132**.

(18) The lateral wall **1413** includes a first lateral wall **1413a** and a second lateral wall **1413b** sequentially distributed from the light source **131** to the light guide plate **132**. The support part **1414** extends from a side of the lateral wall **1413a** away from the support plate **141** toward the accommodating space **150**.

(19) A surface of the support part **1414** away from the support plate **141** is connected to the second display panel **12**.

(20) As illustrated in FIG. **1**, by disposing the support part **1414** extending from the side of the lateral wall **1413a** away from the support plate **141** toward the accommodating space **150**, a U-shaped structure is formed between the support part **1414**, the first lateral wall **1413a**, and the support plate **141**, which allows the light source **131** and the light guide plate **132** can be disposed in the U-shaped structure.

(21) It should be noted that, a gap can be left between a surface of the support part **1414** close to the support plate **141** and the light guide plate **132**, or the surface of the support part **1414** close to the support plate **141** is connected to the light guide plate **132**. Preferably, as illustrated in FIG. **3**, the surface of the support part **1414** close to the support plate **141** is connected to the light guide plate **13** to realize engagement on the light guide plate **132**, which improves stability of the structure.

(22) In one embodiment provided by the present application, as illustrated in FIG. **1**, a connection element **211** is disposed on the support plate **141** close to the second lateral wall **1413b**. An end of the connection element **211** is connected to the second display panel **12**, and another end of the connection element **211** is connected to the support plate **141** to realize support for the second display panel **12**. A plastic material is generally used for the connection element **211**.

(23) It should be noted that at a location of the support plate **141** close to the first lateral wall **1413a** and the second lateral wall **1413b**, a manner of disposing the connection element **211** can be simultaneously adopted to realize support for the second display panel **12**; a manner of bending and extending the lateral wall **1413** to form the support part **1414** is also allowed to be adopted to realize support for the second display panel **12** at the same time; in addition, a manner of disposing the connection element **211** on one side and disposing the support part **1414** on the other side is also allowed to be adopted. When the connection element **211** is adopted, a weight of the entire structure can be reduced. When the support part **1414** is adopted, the rigidity of the structure is relatively great, which has a better support ability for the display panel. Specific embodiments can be determined according to practical situations, which are not limited herein.

(24) As illustrated in FIG. **1**, the first optical film layer **134** of the backlight component **13** is disposed in a box body encircled by the plastic frame **142**, and a gap is left between the edge of the plastic frame **142** and the first optical film layer **134** to leave a space for expansion of the first optical film layer **134**.

(25) In the first embodiment of the present application, as illustrated in FIG. **1**, a via hole **1400** is defined on the support plate **141** to form the light transmissive region **140**. In addition, defining the via hole **1400** can also reduce the weight of the support structure, which is conducive to thinness of the display device **1**. The plastic frame **142** is disposed on a lateral edge of the via hole **1400**.

Therefore, after the first optical film layer **134** expands, the lateral edges of the plastic frame **142** can penetrate the via hole **1400** to connect to the lateral edge of the first optical film layer **134**,

which improves stability of the structure.

(26) FIG. 2 is a structural schematic diagram of the display device of a second embodiment of the present application. As illustrated in FIG. 2, a boss **1410** is disposed on the first lateral surface **1411** of the support plate **141** close to the lateral edge of the via hole **1400**, a groove **1420** corresponding to the boss **1410** is defined on the plastic frame **142**, and the boss **1410** is inserted in the groove **1420**.

(27) The boss **1410** disposed on the lateral edge of the via hole **1400** can serve an effect of enhancing ribs, which improves rigidity and strength of the support plate **141** at the via hole **1400**. By disposing the groove corresponding to the boss **1410** on the plastic frame **142** and by inserting the boss **1410** into the groove, connection between the plastic frame **142** and the support plate **141** is firmer.

(28) It should be noted that, the larger an area of the first display panel **11** is, the larger a required light transmissive region **140** is, and correspondingly, the larger the via hole **1400** that needs to be defined on the support plate **141** is. Preferably, in one embodiment of the present application, the area of the first display panel **11** is less than or equal to an area of the second display panel **12**. Therefore, low strength of the entire structure caused by a large-area via hole **1400** being defined on the support plate **141** is prevented.

(29) It should be noted that, in the first embodiment and the second embodiment of the present application, a plate of stainless steel or a metal material of relative high rigidity such as aluminum of high rigidity can be adopted on the support plate **141**, and the support plate **141** can be formed by a stamping method. The plastic frame **142** and the support plate **141** can be integrally formed. After the support plate **141** is stamped, the plastic frame **142** can be formed on the support plate **141** by a pouring method, thereby simplifying fabrication processes of the support structure **14**.

(30) Regarding the display device **1** in the first embodiment and the second embodiment, when the display device **1** is assembled, the backlight component **13** and the second display panel **12** can be mounted on the support structure **14** in advance, then the support structure **14** is flipped, and then the first display panel **11** is mounted on the support structure **14**. When the backlight component **13** and the second display panel **12** are mounted in advance, the backlight component **13** can provide support for subsequent installation of the first display panel **11** to allow assembling of the display device **1** to be easier.

(31) With reference to FIG. 4 and FIG. 5, the support plate **141** includes the first lateral surface **1411** and the second lateral surface **1412** which are opposite to each other. The support plate **141** has a light transmissive region **140**. The plastic frame **142** and the support plate **141** enclose to form a mounting cavity **200**. The second display panel **12** is disposed in the mounting cavity **200**. The second display panel **12** is located between the first display panel **11** and the first lateral surface **1411** and is attached and connected to the first lateral surface **1411**. A surface of the plastic frame **142** away from the support plate **141** is connected to the first display panel **11**. The plastic frame **142** and the first display panel **11** can be bonded by adhesive tapes to realize fixation of the two.

(32) In a fourth embodiment of the present application, a transparent material such as glass or resin can be adopted on the support plate **141**. The plastic frame **142** can be fixed with the support plate **141** by bonding. The second display panel **12** is located between the first display panel **11** and the first lateral surface **1411** and is connected to the first lateral surface **1411**.

(33) By connecting the second display panel **12** to the support plate **141** and by connecting the surface of the plastic frame **142** away from the support plate **141** to the first display panel **11**, connection between support structure **14** and the display module **10** is realized. Furthermore, support of the support structure **14** for the display module **10** improves the rigidity of the display device **1** and reduces the risk of deformation occurring on the display module **10**.

(34) Because the second display panel **12** is disposed in the mounting cavity **200** enclosed by the plastic frame **12** and the support plate **141**, the larger the size of the second display panel **12** is, the

larger the space of the required mounting cavity **200** is, and the longer a distance d between lateral edges of the plastic frame **142** is. Preferably, the area of the second display panel **12** being less than or equal to the area of the first display panel **11** prevents poor support ability caused by a contacting area between the plastic frame **142** and the first display panel **11** being too small when the distance d between the lateral edges of the plastic frame **142** is too long.

(35) Regarding the display device **1** in the fourth embodiment, when the display device **1** is assembled, the second display panel **12** can be bonded to the support plate **141** first, then the plastic frame **142** is disposed on the support plate **141** to allow the plastic frame **142** and the support plate **141** to encircle to form the mounting cavity **200**, and then the second optical film layer **135** is put into the mounting cavity **200**. In this way, a film structure can be prevented from scattering when the second optical film layer **135** is assembled first.

(36) After the second display panel **12** is assembled, the backlight component **13** and the first display panel **11** is mounted on the support structure **14**. Finally, an encapsulation adhesive is attached (not shown in the figure).

(37) It should be noted that, a gap can be left between a lateral edge of the backlight component **13** and a lateral edge of the plastic frame **142**, or the backlight component **13** and the plastic frame **142** can be connected to each other. Preferably, in the fifth embodiment of the present application, as illustrated in FIG. **6**, the lateral edge of the plastic frame **142** is connected to a lateral edge of the light guide plate **132** in the backlight component **13** to realize engagement for the backlight component **13** to improve stability of the structure (the light source in the backlight component **13** and the transfective film are not shown in FIG. **4** and FIG. **6**).

(38) The display device provided by embodiments of present disclosure are described in detail above. This article uses specific cases for describing the principles and the embodiments of the present disclosure, and the description of the embodiments mentioned above is only for helping to understand the method and the core idea of the present disclosure. It should be understood by those skilled in the art, that it can perform changes in the technical solution of the embodiments mentioned above, or can perform equivalent replacements in part of technical characteristics, and the changes or replacements do not make the essence of the corresponding technical solution depart from the scope of the technical solution of each embodiment of the present disclosure.

Claims

1. A display device, comprising: a display module comprising a first display panel and a second display panel disposed oppositely and a backlight component disposed between the first display panel and the second display panel, wherein the backlight component comprises a light guide plate and a light source disposed on a lateral side of the light guide plate, the first display panel and the second display panel have light incident surfaces, and the light incident surfaces of the first display panel and the second display panel are respectively opposite to the backlight component; and a support structure, wherein the support structure is connected to the display module and supports the display module, the support structure comprises a support plate and a plastic frame, the support plate comprises a first lateral surface and a second lateral surface which are opposite to each other, the support plate has a light transmissive region, the plastic frame is protrusively disposed on the first lateral surface, the plastic frame is distributed at a periphery of the light transmissive region, and a surface of the plastic frame away from the support plate is connected to the first display panel, wherein a via hole is defined on the support plate to form the light transmissive region, and the plastic frame is disposed on a lateral edge of the via hole, and a boss is disposed on the first lateral surface close to the lateral edge of the via hole, a groove corresponding to the boss is defined on the plastic frame, and the boss is inserted in the groove.

2. The display device as claimed in claim 1, wherein the support structure further comprises a lateral wall protrusively disposed on a lateral edge of the support plate, the support plate and the

- lateral wall enclose an accommodating space, the backlight component is disposed in the accommodating space; a support part extends from a side of the lateral wall away from the support plate toward the accommodating space, and a surface of the support part away from the support plate is connected to the second display panel.
3. The display device as claimed in claim 2, wherein the lateral wall comprises a first lateral wall and a second lateral wall sequentially distributed from the light source to the light guide plate, the support part extends from a side of the first lateral wall away from the support plate toward the accommodating space.
4. The display device as claimed in claim 3, wherein a surface of the support part close to the support plate is connected to the light guide plate.
5. The display device as claimed in claim 1, wherein a material of the support plate is stainless steel or aluminum.
6. The display device as claimed in claim 5, wherein the support plate and the plastic frame are integrally formed.
7. A display device, comprising: a display module comprising a first display panel and a second display panel disposed oppositely and a backlight component disposed between the first display panel and the second display panel, the first display panel and the second display panel have light incident surfaces, and the light incident surfaces of the first display panel and the second display panel are respectively opposite to the backlight component; and a support structure, wherein the support structure is connected to the display module and supports the display module, the support structure comprises a support plate and a plastic frame, the support plate comprises a first lateral surface and a second lateral surface which are opposite to each other, the support plate has a light transmissive region, the plastic frame is protrusively disposed on the first lateral surface, the plastic frame is distributed at a periphery of the light transmissive region, and a surface of the plastic frame away from the support plate is connected to the first display panel, wherein a via hole is defined on the support plate to form the light transmissive region, and the plastic frame is disposed on a lateral edge of the via hole, and a boss is disposed on the first lateral surface close to the lateral edge of the via hole, a groove corresponding to the boss is defined on the plastic frame, and the boss is inserted in the groove.
8. The display device as claimed in claim 7, wherein the support structure further comprises a lateral wall protrusively disposed on a lateral edge of the support plate, the support plate and the lateral wall enclose an accommodating space, the backlight component is disposed in the accommodating space; a support part extends from a side of the lateral wall away from the support plate toward the accommodating space, and a surface of the support part away from the support plate is connected to the second display panel.
9. The display device as claimed in claim 8, wherein the backlight component comprises a light guide plate and a light source disposed on a side of the light guide plate, the lateral wall comprises a first lateral wall and a second lateral wall sequentially distributed from the light source to the light guide plate, the support part extends from a side of the first lateral wall away from the support plate toward the accommodating space.
10. The display device as claimed in claim 9, wherein a surface of the support part close to the support plate is connected to the light guide plate.
11. The display device as claimed in claim 9, wherein a connection element is disposed on the support plate close to the second lateral wall, and an end of the connection element is connected to the second display panel, and another end of the connection element is connected to the support plate.
12. The display device as claimed in claim 8, wherein an area of the first display panel is less than or equal to an area of the second display panel.
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